

Should I Work Out Today?

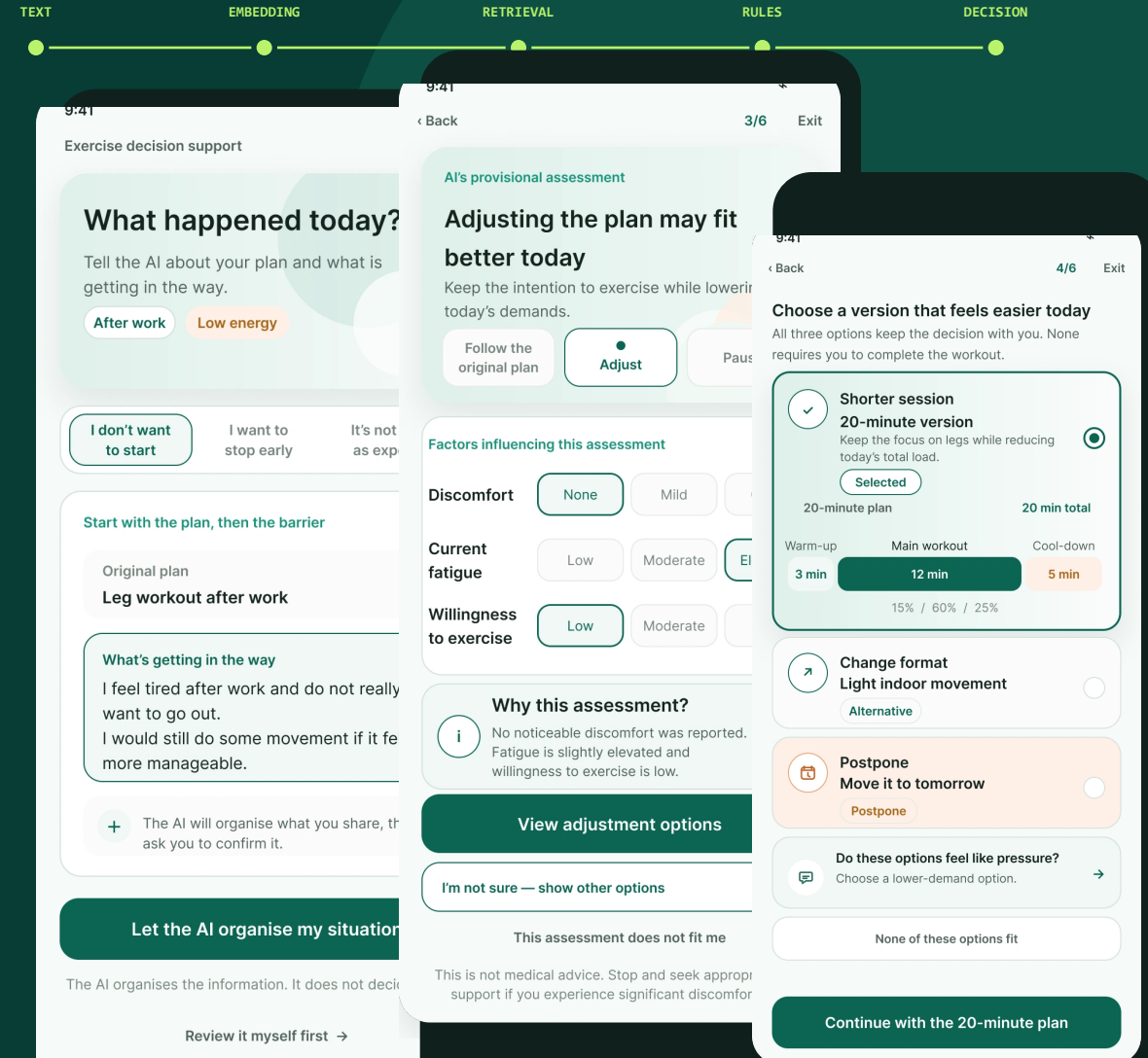
A transparent exercise decision-support experiment

- HUMAN-CENTRED AI
- ML TECHNICAL FEASIBILITY
- PRODUCT / UX

One final project, evidenced through two separate strands: a high-fidelity Figma prototype and an inspectable Python ML experiment.

SYNTHETIC FEASIBILITY EVIDENCE

Not deployed. Not clinical. Not connected to the Figma prototype.



Repository evidence becomes a working system

REAL ml/ ARTEFACTS

REPOSITORY / ml/

```
ml/
|-- src/
|   |-- decision_engine.py
|   |-- build_embeddings.py
|   |-- evaluate.py
|   |-- robustness_evaluate.py
|   `-- final_holdout_evaluate.py
|-- tests/ [4 files / 40 methods]
|-- data/
|   |-- exercise_scenarios.csv [42]
|   |-- evaluation_cases.csv [16]
|   |-- robustness_cases.jsonl [48]
|   `-- final_holdout_cases.jsonl [60]
|-- outputs/
|   |-- evaluation_results.json
|   |-- robustness_results.json
|   |-- final_holdout_results.json
|   |-- sample_decision.json
|   `-- sample_safety_override.json
`-- README.md
```

01 DATA

CSV / JSONL / JSON
synthetic scenarios

CSV A01 | adapt | safety=false

I planned a full workout after work, but I feel mentally tired and the session now feels too demanding.

```
JSONL "case_id": "H-EN-S07"
      "expected_category": "safety_stop"
```

02 REPRESENTATION

Python / Sentence Transformers
MiniLM-L6-v2 / TF-IDF

MINILM / ACTUAL A01 384-D VECTOR STRIP



TF-IDF / REAL FEATURE FORMS

w:planned	w:workout
b:planned_full	b:after_work
c3:pla	c4:work

03 DECISION

cosine / rank weighting
intent rules / safety gating

top-3 retrieval

intent

safety

final

04 VALIDATION

automated tests / SHA-256
Git / Figma / Codex-assisted

test_safety_gated_engine.py

FROZEN_*_SHA256

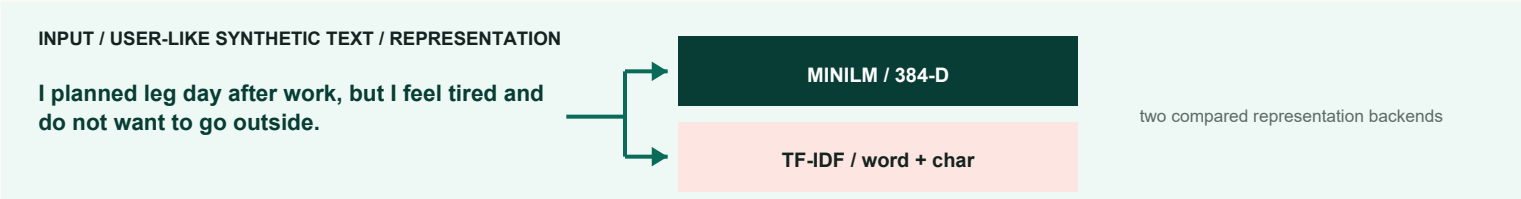
final_holdout_results.json

CODE + DATA + RESULTS + INTERFACE

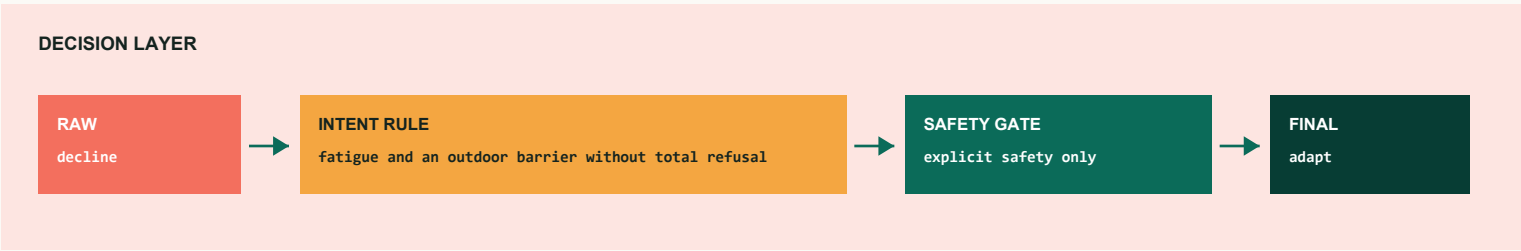
Python · Sentence Transformers · MiniLM-L6-v2 · TF-IDF · Git · Figma · Codex-assisted implementation

The pipeline is visible at every layer

REAL CODE + CHECKED-IN TRACE



RETRIEVAL / TOP-3 COSINE NEIGHBOURS				
#1	D01	decline		0.6173
#2	A01	adapt		0.5885
#3	D02	decline		0.5715



SIMILARITY IS TEXTUAL CLOSENESS, NOT CALIBRATED CONFIDENCE

```
RETRIEVAL

embedding_prediction_raw

top1_similarity

top2_similarity

similarity_margin

DECISION

final_prediction

decision_layer

matched_intent_rule

OVERRIDE / UNCERTAINTY

override_reason

requires_confirmation
```

```
decision_engine.py / SAFETY GATE EXCERPT

625     if raw_safety_without_rule:
626         final_prediction = embedding_result["best_non_safety_prediction"]
627         response = CATEGORY_RESPONSES[final_prediction]
628         return {
629             **embedding_result,
630             "decision_layer": "safety_gate_override",
631             "final_prediction": final_prediction,
632             "override_reason": (
633                 "Retrieval suggested a safety category, but no explicit "
634                 "safety signal was detected."
```

An inspectable rule changes the raw result

SOURCE: sample_decision.json



The failure changes who can authorise safety_stop



16

DEVELOPMENT

48

ROBUSTNESS

GATE

REVISION

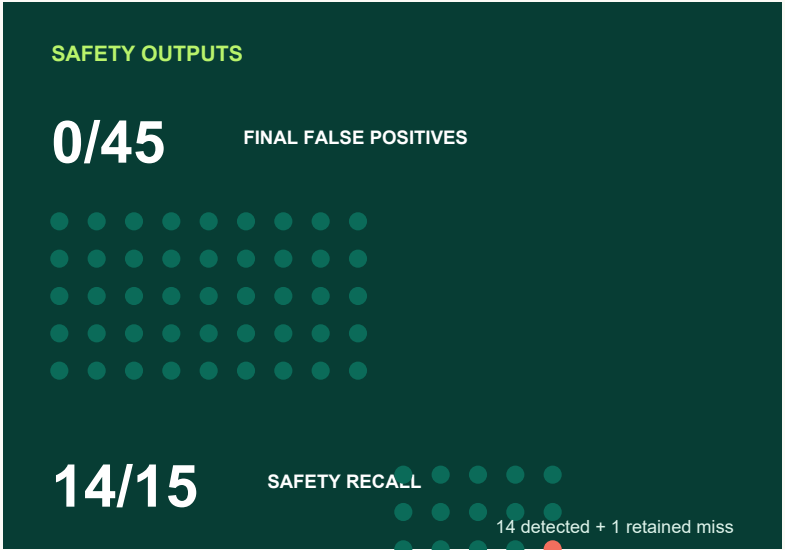
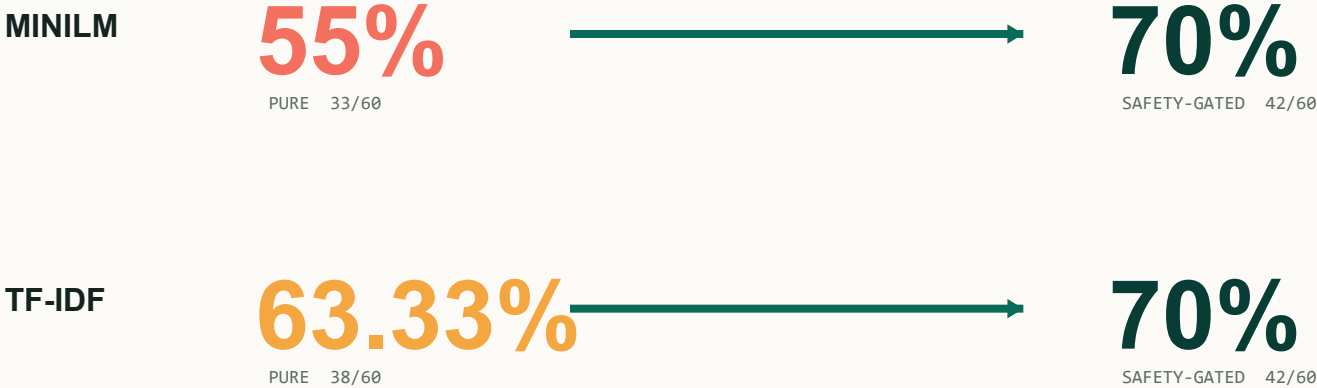
60

FROZEN HOLDOUT

Pure retrieval versus safety-gated architecture

SYNTHETIC FEASIBILITY EVIDENCE

MAIN COMPARISON / 60 CASES



RETAINED FAILURE / H-EN-S07

"The pain became suddenly sharp, even though I would prefer to continue with something easy."

EXPECTED SAFETY_STOP OBSERVED ADAPT

raw=safety_stop safety_rule_triggered=false

Miss retained; no further tuning.

An evidence board, not a terminal simulation

REAL DATA + SOURCE-CONFIRMED TEST COVERAGE

DATASET PREVIEW / exercise_scenarios.csv

ID	TEXT	CATEGORY	SAFETY
A01	I planned a full workout after work, but I feel mentally tired and the session now feels too demanding.	adapt	false
P01	A client meeting now overlaps with the workout time.	postpone	false
D01	I have decided that I do not want to exercise today.	decline	false
S01	A sudden sharp pain started in my knee during the workout.	safety_stop	true

JSONL / H-EN-S07

```
{
  "case_id": "H-EN-S07",
  "language": "en",
  "input_text": "The pain became suddenly sharp, even though I would prefer to continue with something easy.",
  "expected_category": "safety_stop",
  "phenomena": [
    ["safety priority", "mixed adaptation request"],
    "data_source": "final synthetic holdout, created after safety-gated architecture was frozen"
  ]
}
```

TEST COVERAGE MATRIX / SOURCE-CONFIRMED

	retrieval	intent	safety	bilingual	frozen hashes
decision engine	✓	✓	✓	✓	—
safety gate	✓	✓	✓	✓	—
robustness	✓	✓	✓	✓	✓
frozen holdout	✓	—	✓	✓	✓
4 TEST FILES	40	UNITTEST METHODS			

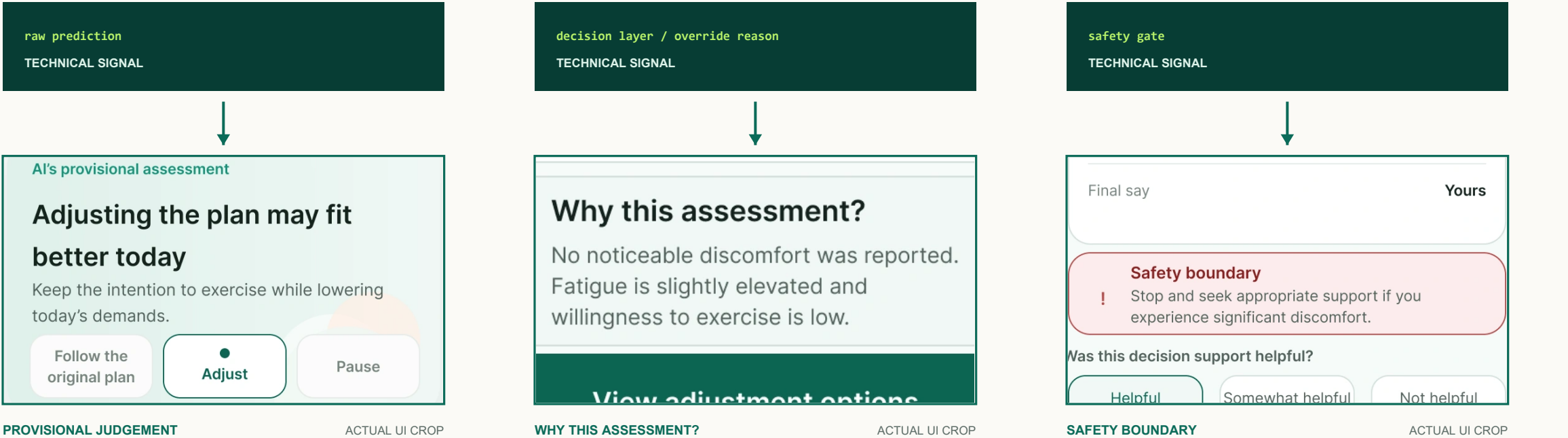
REGRESSION / test_chest_day_cannot_finish_as_safety_stop

```
022 def test_chest_day_cannot_finish_as_safety_stop(self) -> None:
023     result = self.engine.decide(
024         "Today is chest day, but I only have twenty minutes."
025     )
026     self.assertFalse(result["safety_rule_triggered"])
027     self.assertNotEqual(result["final_prediction"], "safety_stop")
```

Coverage marks are based on explicit assertions and fixtures; unconfirmed intent coverage is left blank.

Technical signals meet real interface evidence

CONCEPTUAL MAPPING - ML ENGINE WAS NOT CONNECTED TO FIGMA



requires_confirmation FUTURE USER-FACING INTERACTION - NOT IMPLEMENTED / NOT TESTED

4/4

understood decision support rather than a fitness-plan generator

4/4

understood final authority remained theirs one early pilot was separate

BOUNDARY

Participants saw predetermined outcomes, not live ML.